Fig. 4: High-resolution CIJ printers can print as small as 0.6mm in height

a laser beam that alters the surface of the component and ensures establishment of permanent, traceable marks—from simple codes to more complex graphs. As these printers do not use inks, codes are not affected by the alcohol cleaning processes. Manufacturers have the option of several laser coding methods, depending on the sensitivity of components. New advanced printer ranges offer a variety of laser sources, including CO_2, fibre, UV and YAG.

In order to safeguard the quality of codes, manufacturers can use integrated vision systems to verify the accuracy of codes and ensure these are legible for supply chain and distribution partners. These systems can identify whether the contrast in coding is adequate, and help to highlight common coding errors.

Conclusion

Counterfeiting has the potential to create significant profits or losses for companies. This is a real and present issue within the electrical component industry, which can be addressed, in part, through coding and marking applications.

Component manufacturers must also deal with the added challenge of printing increasingly complex codes on small substrate surfaces, and countering these issues with outdated technologies is not a sustainable option. Selecting an advanced CIJ or laser printer with smart coding capabilities can go a long way in assisting manufacturers tackle the problem of counterfeiting, at the same time ensuring maximised production line performance. EFY